Machine Learning

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Objectives

- understand and explain the basic concepts of machine learning
- understand formalized concepts and methods

- be able to implement concepts and methods in the form of algorithms
- be able to sensibly select, adapt, and apply relevant methods
- be able to educate oneself

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Related Fields

1. Statistics [paradigms, models]
2. Mathematics

3. Information Retrieval [methods, algorithms]
4. Knowledge Processing
5. Heuristic Search

6. Decision Support Systems [applications]
7. Artificial Intelligence
8. Web Technology

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Literature

Machine Learning:

- C.M. Bishop. [Interview (2018 MS Research)]
  Pattern Recognition and Machine Learning
  2nd edition, Springer 2006.

- L. Breiman, J.H. Friedman, R.A. Olshen, C.J. Stone.
  Classification and Regression Trees
  CRC Press reprint, 1998.

- N. Cristianini, J. Shawe-Taylor.
  An Introduction to Support Vector Machines and Other Kernel-based Learning Methods
  Cambridge University Press, 2000.

- T. Hastie, R. Tibshirani, J. Friedman.
  The Elements of Statistical Learning
  2nd edition, Springer, 2009. statweb.stanford.edu/∼hastie/ElemStatLearn/

- T. Mitchell.
  Machine Learning
  1st edition, McGraw-Hill, 1997. www.cs.cmu.edu/∼tom/mlbook.html

- V. Vapnik.
  The Nature of Statistical Learning Theory
  2nd edition, Springer 2000.

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Software

Programming:

- Eclipse Foundation, Inc., Canada.
  Eclipse SDK
  Version 4.13. www.eclipse.org/downloads

Machine Learning:

- E. Frank, M. Hall, G. Holmes, M. Mayo, B. Pfahringer, T. Smith, I. Witten.
  Weka Machine Learning Project
  Version 3.8. www.cs.waikato.ac.nz/ml/weka

- *scikit-learn: Machine Learning in Python*
  Version 0.21 http://scikit-learn.org/stable/

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Software

Statistics:

- R Development Core Team.
  R
  Version 3.6. www.r-project.org

- E. Jones, T. Oliphant, P. Peterson and others.
  SciPy
  Version 1.17. www.scipy.org

- J.W. Eaton.
  GNU Octave
  Version 5.1. www.gnu.org/software/octave

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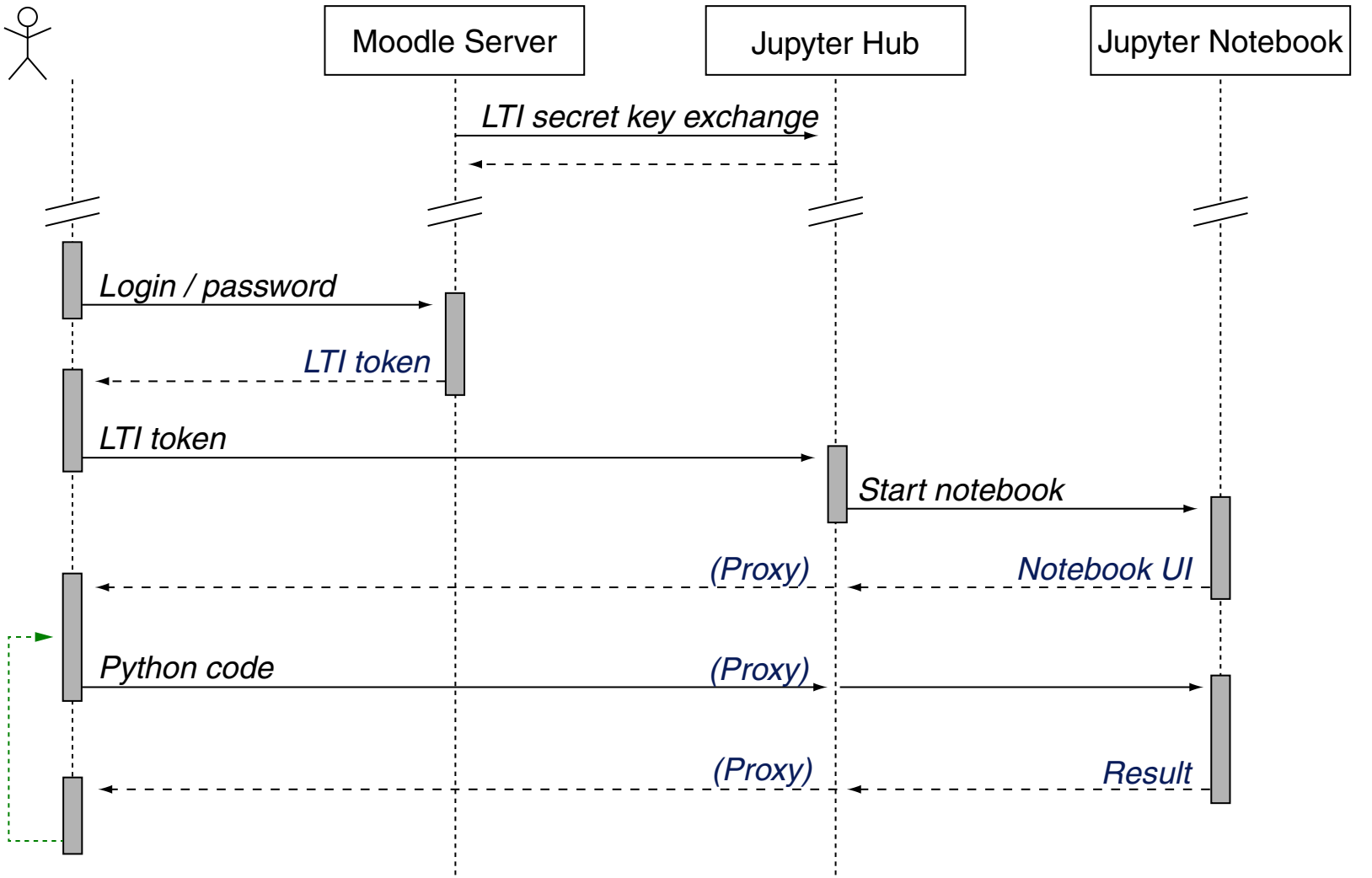
Software

Lab Class Setup



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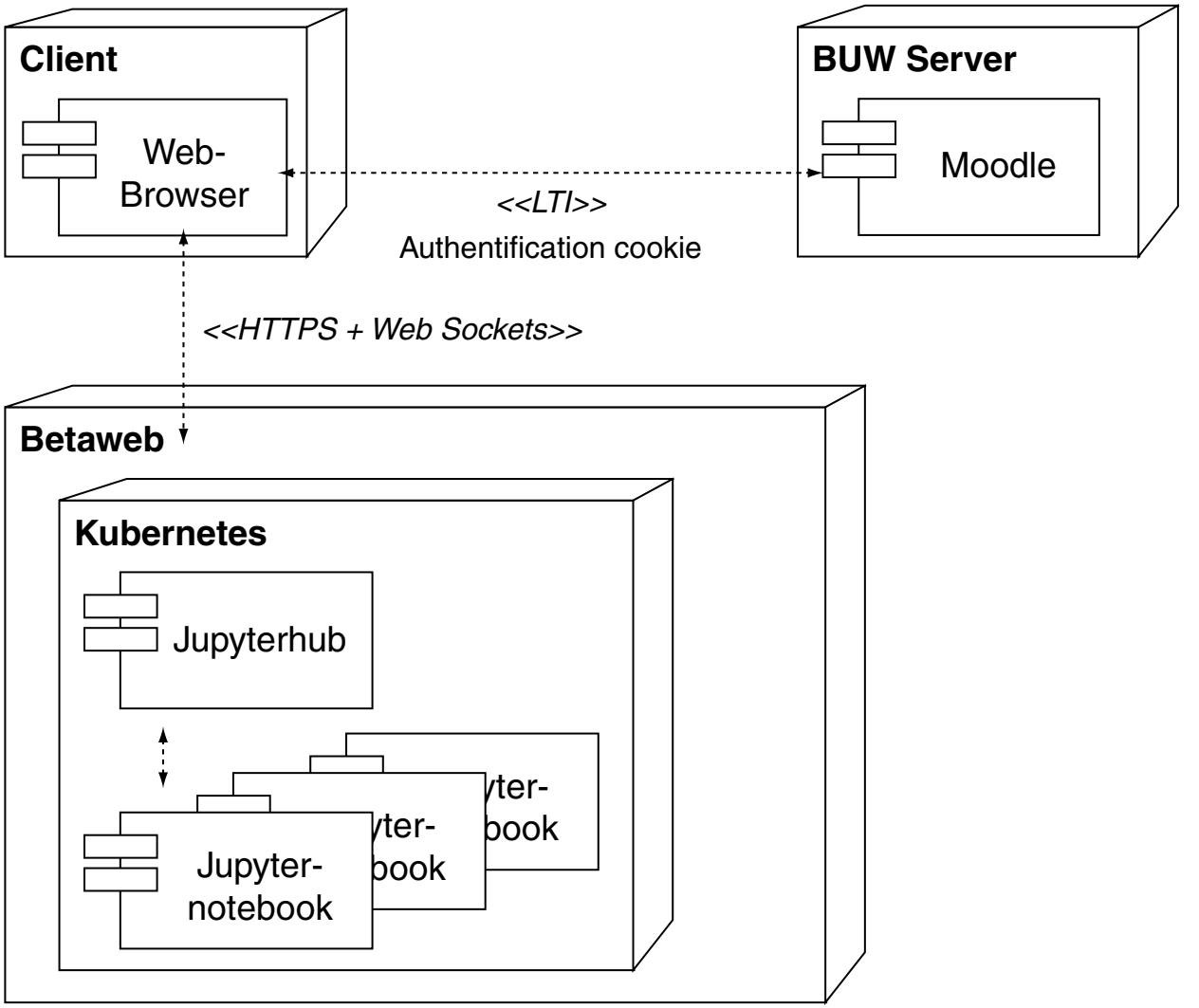
Software

Lab Class Setup (continued)



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